

Celestial coordinate systems

Reference handout and exercises

Coordinate systems describe directions and positions. Astronomers choose a system that fits the observation or question.

The celestial sphere

Imagine distant objects projected onto a sphere centered on the observer. This model describes directions without representing the objects' distances.

Earth's rotation makes stars appear to move across the local sky. Different coordinate systems describe this motion in different ways.

1 Spherical coordinates

A spherical coordinate system uses a radius and two angles. The diagram shows one common convention for r , θ , and ϕ .

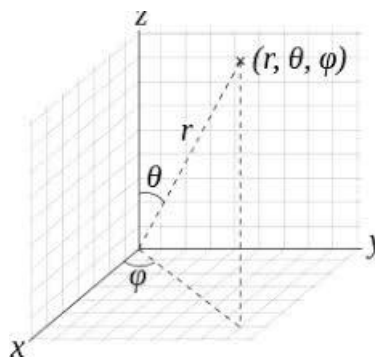


Figure 1. Radius and angles in a spherical coordinate system.

2 Longitude and latitude on Earth

Latitude measures angle north or south of the equator. Longitude measures angle east or west of the prime meridian.

The prime meridian passes through Greenwich and defines 0° longitude.

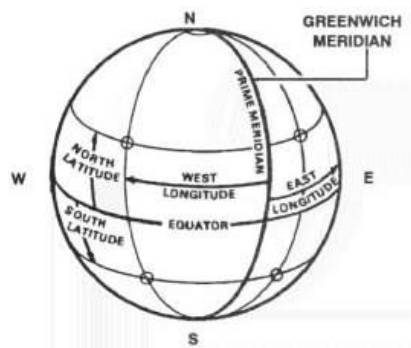


Figure 2. Terrestrial latitude and longitude.

Horizontal coordinates

Horizontal coordinates describe the sky from a particular place and time. A star's horizontal coordinates change as Earth rotates.

Term	Meaning
Zenith	The direction straight overhead, at 90° elevation
Nadir	The direction opposite the zenith, straight down
Horizon	The reference plane perpendicular to the zenith–nadir direction
Celestial meridian	The great circle through the celestial poles, zenith, and nadir
Azimuth	Direction around the horizon, measured from north through east
Elevation	Angle above the horizon; also called altitude

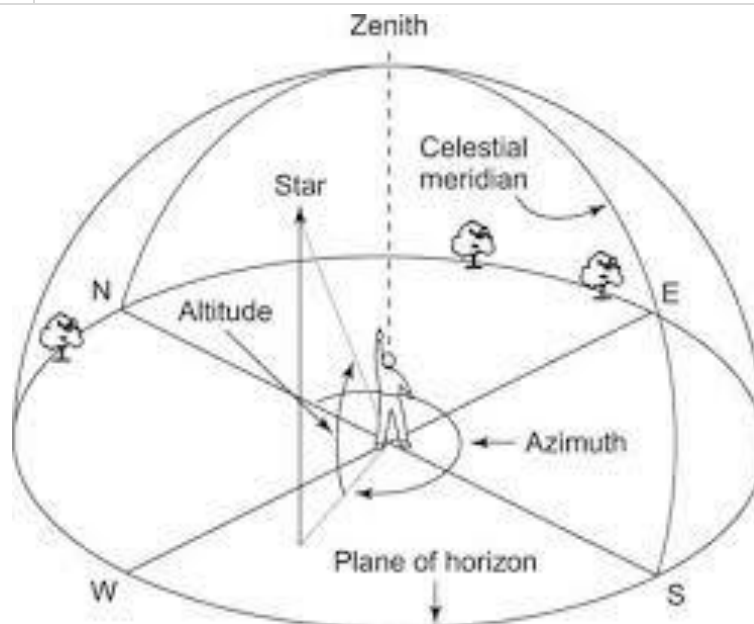


Figure 3. Azimuth and elevation locate an object in the local sky.

In this convention, north is 0° , east is 90° , south is 180° , and west is 270° azimuth.

Equatorial coordinates

Right ascension and declination provide sky coordinates that do not depend on the observer's location.

For a short observing session, a distant star's catalog coordinates remain effectively fixed. Its azimuth and elevation still change.

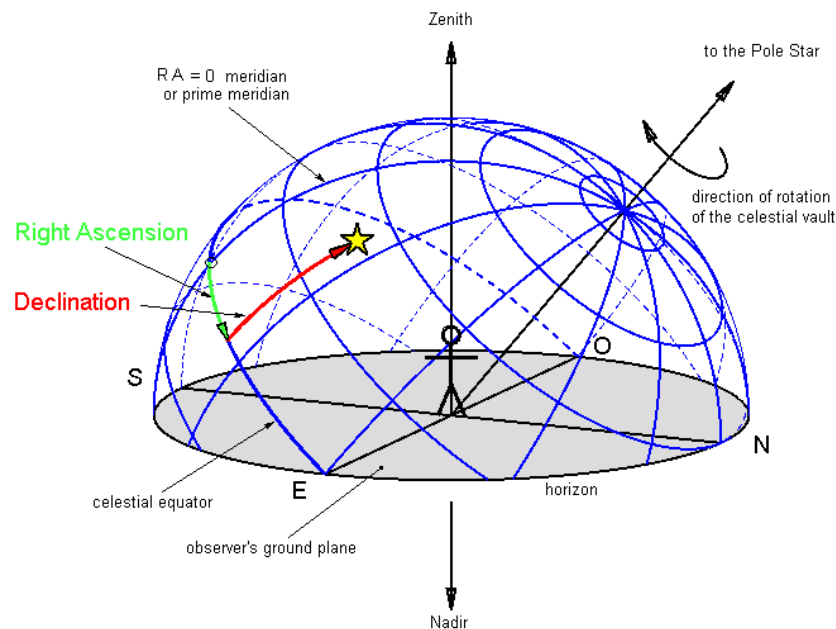


Figure 4. Right ascension and declination on the celestial sphere.

Right ascension (RA)

RA measures angle eastward along the celestial equator from the March equinox. One hour of RA equals 15° .

Declination (Dec)

Declination measures angle north or south of the celestial equator. The equator has declination 0° .

The north celestial pole has declination $+90^\circ$. The south celestial pole has declination -90° .

The celestial equator projects Earth's equator onto the sky. The celestial poles lie along Earth's rotation axis.

Galactic coordinates and exercises

Galactic coordinates describe directions relative to the Milky Way's plane, as viewed from our position in the Galaxy.

Coordinate	Definition
Galactic longitude	Angle around the Galactic plane, from 0° toward the Galactic center to 360°
Galactic latitude	Angle above or below the Galactic plane, from -90° to $+90^\circ$

Longitude 180° points away from the Galactic center. The original diagram divides longitude into four quadrants.

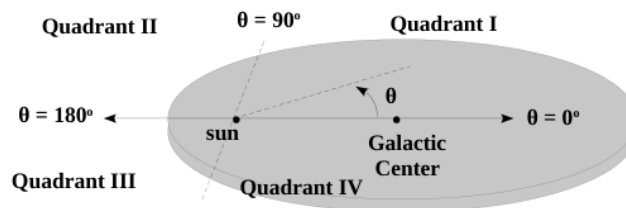


Figure 5. Galactic longitude and quadrants viewed from above the Galactic plane.

Exercises

1. An observer in Charlotte looks straight up. How does that direction's declination relate to the observer's latitude?
2. A star rises in the east at 30° elevation, then passes overhead. Does its RA change during this motion? Explain.

Use a sketch to distinguish movement across the local sky from a change in a star's catalog coordinates.

Sources

Adapted from the DSPIRA coordinate-systems handout. Original diagrams retained.

[NASA JPL: Right ascension](#)

[NASA: Using multiwavelength Milky Way maps](#)